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Study of different types of network cables.

Aim:

To study different types of network cables.

a) understand different types of network cable:

Different types of cables used in networking are:

i) unshielded twisted Pair (UTP) cable

ii) shielded twisted Pair (STP) cable.

iii) Coaxial cable.

iv) Fibre optic cable.

Cable type	Category	Maximum Data transmi	Advantages / Disadvantages	Application / use.
UTP	category 3	10 bps	Advantages: i) cheaper in cost ii) easy to install Disadvantages: i) More prone to (EMI) Electromargetic interface & noise.	i) 10 Base-T Ethernet
	category 5	upto 100 Mbps		ii) Fast Ethernet, Gigabit Ethernet
	category 5e	1 Gbps		iii) Fast Ethernet gigabit Ethernet

b) Make your own Ethernet Cross-over Cable / straight Cable:

Tools & parts needed:

i) Ethernet cabling. Caste is certified for gigabit support, but CAT5 cabling works on well, just over shorting distances.

ii) A crimping tool. this is an all-in-one networking tool shaped to push down the pins & plug & strip and cut the shielding off the cables.

iii) Two RJ45 plug.

iv) optional two plug shields.

Step 1: To start construction of the device, begin by threading shield onto the cable.

Step 2: Next, strip approximately 1.5 cm of cable shielding from both ends. The crimping tool has a round area to complete this task

Step 3: After, you will need to untangle the wires; there should be four "twisted pairs" referencing back to the sheet, arrange them

Cable type	Category	Maximum Data Transmission	Advantages / Disadvantages	Application / Use.
STP	Category 6, 6a	10 Gbps	Advantages: i) shielded ii) Faster than UTP Disadvantages: i) Expensive ii) Greater in installation effort	Gigabit Ethernet, 10G, Ethernet (55 m) widely used in data center.
SSTP	Category	10 Gbps		Gigabit Ethernet, 10G Ethernet (100 m)
Coaxial Cable	RG-6, RG-59 RG-11	10 - 100 Mbps	Advantages: i) High bandwidth ii) Versatile Disadvantages: i) cost ii) Size is bulky	Speed of signal is 500m Television network high speed internet connection
fibre optics Cable	Single mode Multi mode	100 Gbps	Advantages: i) High speed ii) High bandwidth iii) High Security iv) Long Distance Disadvantages: i) Expensive ii) Requires Skills installed	Maximum distance of fibre optics cable is around 100 meters.

fragile plastic plug.

Step 7: Lastly, repeat for the other end using diagram B (to make a crossover cables) / using diagram A (to make a straight through cable).

Result:

Thus the different types of network cables are studied

from top to bottom. One end should be in arrangement A and the other in B.

Step 4: Once the order is correct, bunch them together in a line and if there are any that stick them back to create an even level. The difficult aspect is placing these into the RJ45 plug without messing up the order. To do so, hold the plug with the clip side facing away from you and have the gold pins facing towards you as shown.

Step 5: Next push the cable right in. The notch at the end of the plug needs to be just over the cable shielding. simply slip the cables back a little more.

Step 6: After the wires are securely sitting inside the plug, insert it into the crimping tool & push down.

⇒ It should be shaped correctly, but pushing too hard can crack the